

Exp no:10

Implement the Boundary condition between two different dielectric media in Magnetic Field and find the relative permeability of the different medium.

Aim:

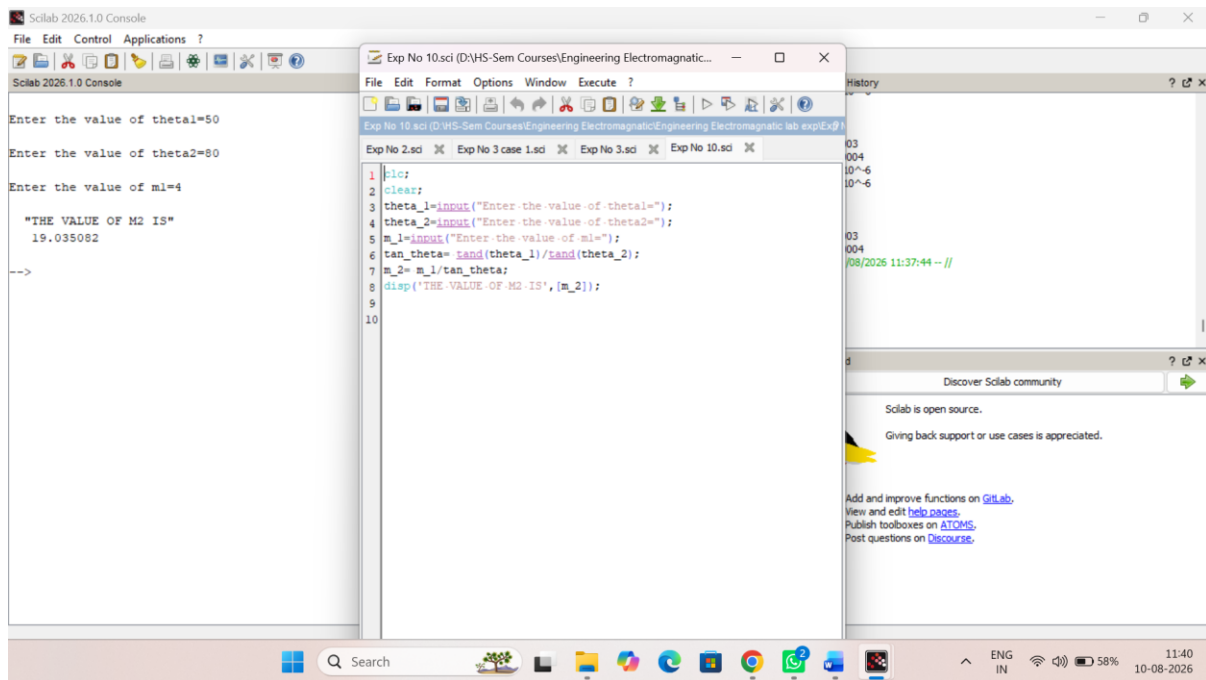
To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB.

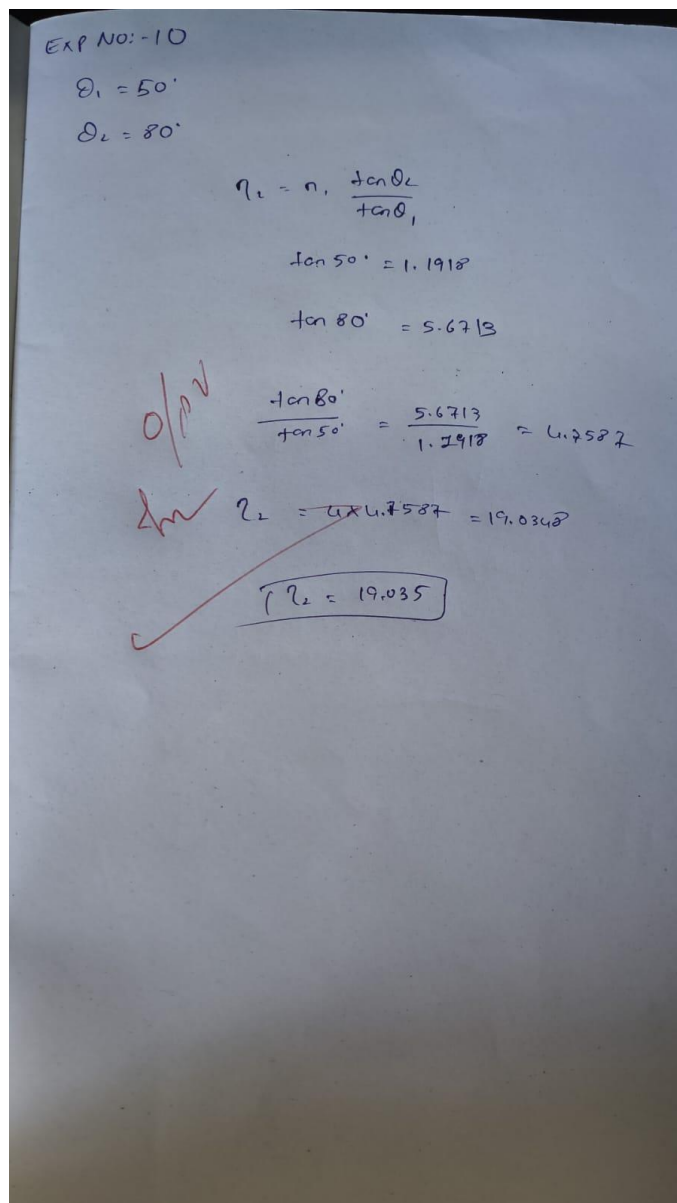
Software required:

- SCILAB version 6.0.1

Theory:

When a magnetic field is present at the boundary between two different dielectric media (or magnetic materials), boundary conditions govern how the magnetic field behaves as it crosses from one medium to the other. These boundary conditions arise from Maxwell's equations and the properties of the materials involved, such as relative permeability, μ_r , which governs the material's response to the magnetic field.





Result :

Thus the program was verified for Boundary condition between two different dielectric media in magnetic field using SCILAB was successfully.

Parameters	Simulated Result	Theoretical result
Theta 1 = 50 Theta 2 = 80 M1 =4	19.035	19.035

Case 1: Consider two identical media in the presence of a magnetic field. The relative permeability in both media is 1. What will be the effect of this on the magnetic field and flux density across the boundary?

The screenshot shows the Scilab 2026.1.0 interface. The console on the left displays the execution of a script, showing the magnetic field intensity and flux density in two media. The editor on the right shows the source code for 'Exp No 10 case 1.sci', which simulates the magnetic field and flux density for two identical media with relative permeability of 1.

```

// Scilab code to simulate the magnetic field and flux density for two identical media
// Constants
mu_0=4*pi*10^(-7); // Permeability of free space (H/m)
H1=10; // Magnetic field intensity in medium 1 (A/m)
// Relative permeability for both media (both are 1)
mu_r1=1; // Relative permeability for medium 1
mu_r2=1; // Relative permeability for medium 2
// Calculate the magnetic field intensity in both media (no change because mu_r = 1)
H2=H1; // Since the media are identical, H1 = H2
// Magnetic flux density in both media (B = mu * H)
B1=mu_0*mu_r1*H1; // Magnetic flux density in medium 1
B2=mu_0*mu_r2*H2; // Magnetic flux density in medium 2
// Display results
disp("Magnetic Field Intensity in Medium 1: "+string(H1)+" A/m");
disp("Magnetic Field Intensity in Medium 2: "+string(H2)+" A/m");
disp("Magnetic Flux Density in Medium 1: "+string(B1)+" T");
disp("Magnetic Flux Density in Medium 2: "+string(B2)+" T");
  
```

Result : Thus, the effect of two identical media on the magnetic field and flux density across the boundary is calculated.

Parameters	Simulated Result	Theoretical result
1. Magnetic Field Intensity in Medium 1: 10 A/m	Magnetic Flux Density in Medium 1: 0.0000126 T	Magnetic Flux Density in Medium 1: 0.0000126 T
2. Magnetic Field Intensity in Medium 2: 10 A/m	Magnetic Flux Density in Medium 2: 0.0000126 T	Magnetic Flux Density in Medium 2: 0.0000126 T

Case 2: In a system with two media, where the relative permeability of medium 1 is $\mu_{r1}=4$ and medium 2 is a vacuum ($\mu_{r2}=1$), how would the magnetic field intensity and flux density behave at the boundary?

Analytical Calculation:

Given:

1. Medium 1 has relative permeability $\mu_{r1} = 4$.
2. Medium 2 is a vacuum with relative permeability $\mu_{r2} = 1$.
3. Magnetic field intensity in medium 1: $H_1 = 10 \text{ A/m}$.
4. We need to determine how **H** and **B** behave at the boundary between the two media.

Properties:

1. Permeability of medium 1:

$$\mu_1 = \mu_0 \mu_{r1} = 4\mu_0$$

2. Permeability of medium 2 (vacuum):

$$\mu_2 = \mu_0$$

3. Magnetic flux density (**B**) is related to **H** by:

$$\mathbf{B} = \mu \mathbf{H}$$

Boundary Conditions:

1. Tangential component of **H**:

$$H_{1t} = H_{2t}$$

The tangential components of **H** are continuous across the boundary.

2. Normal component of **B**:

$$B_{1n} = B_{2n}$$

The normal components of **B** are continuous across the boundary.

Step 1: Magnetic Flux Density (B_1 in Medium 1)

In medium 1:

$$B_1 = \mu_1 H_1 = (4\mu_0) \cdot 10 = 40\mu_0 \text{ T}.$$

Step 2: Tangential Component of H in Medium 2 (H_2)

Since the tangential components of **H** are continuous across the boundary:

$$H_2 = H_1 = 10 \text{ A/m}.$$

Step 3: Magnetic Flux Density (B_2 in Medium 2)

In medium 2:

$$B_2 = \mu_2 H_2 = \mu_0 \cdot 10 = 10\mu_0 \text{ T.}$$

Step 4: Normal Component of B

The normal components of $\mathbf{B}$ are continuous:

$$B_{1n} = B_{2n}.$$

From the values calculated:

- $B_1 = 40\mu_0 \text{ T}$
- $B_2 = 10\mu_0 \text{ T}$

For B to be continuous, the angle of the magnetic field vectors at the boundary would adjust such that the normal components of B are the same.

The screenshot displays the Scilab 2026.1.0 interface. The console window on the left shows the execution of a script that calculates magnetic field intensity and flux density in two media. The editor window on the right shows the source code for 'Exp No 10 case 2.sci'. The code defines constants for permeability, calculates the magnetic field intensity in medium 2 based on the boundary condition, and then calculates the magnetic flux density in both media. The results are displayed in the console.

```
// Scilab code to simulate the magnetic field intensity and flux density across the boundary
// Constants
mu_0=4*pi*10^(-7); // Permeability of free space (H/m)
H1=10; // Magnetic field intensity in medium 1 (A/m)
// Relative permeability for the media
mu_r1=4; // Relative permeability for medium 1
mu_r2=1; // Relative permeability for medium 2 (vacuum)
// Calculate the magnetic field intensity in medium 2 using the boundary condition
H2=mu_r1*H1/mu_r2; // H2 = (mu_r1 * H1) / mu_r2
// Calculate the magnetic flux density in both media (B = mu * H)
B1=mu_0*mu_r1*H1; // Magnetic flux density in medium 1
B2=mu_0*mu_r2*H2; // Magnetic flux density in medium 2
// Display results
disp("Magnetic Field Intensity in Medium 1: "+string(H1)+" A/m");
disp("Magnetic Field Intensity in Medium 2: "+string(H2)+" A/m");
disp("Magnetic Flux Density in Medium 1: "+string(B1)+" T");
disp("Magnetic Flux Density in Medium 2: "+string(B2)+" T");
```

Result :

Thus the effect of two different media on the magnetic field and flux density across the boundary is calculated.

Parameters	Simulated Result	Theoretical result
1. Magnetic Field Intensity in Medium 1: 10 A/m	Magnetic Flux Density in Medium 1: 0.0000503 T	Magnetic Flux Density in Medium 1: 0.0000503 T

2. Magnetic Field Intensity in Medium 2: 40 A/m	Magnetic Flux Density in Medium 2: 0.0000503 T	Magnetic Flux Density in Medium 2: 0.0000503 T
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Case 3: If medium 1 has a relative permeability $\mu_{r1}=0.3$ and medium 2 has a relative permeability $\mu_{r2}=2$, calculate the magnetic field intensity in both media at the boundary if the magnetic field intensity in medium 1 is known to be $H_1=10$ A/m.

Analytical Calculations:

Given:

1. Relative permeability:

- Medium 1: $\mu_{r1} = 0.3$
- Medium 2: $\mu_{r2} = 2$

2. Magnetic field intensity in medium 1: $H_1 = 10$ A/m.

3. The permeability in each medium is:

$$\mu_1 = \mu_0 \mu_{r1}, \quad \mu_2 = \mu_0 \mu_{r2}.$$

4. Relationship between **B** and **H**:

$$\mathbf{B} = \mu \mathbf{H}.$$

Step 1: Calculate B_1 in Medium 1

$$B_1 = \mu_1 H_1 = (\mu_0 \mu_{r1}) H_1 = (\mu_0 \cdot 0.3) \cdot 10 = 3\mu_0 \text{ T}.$$

Step 2: Tangential Component of H in Medium 2

Since the tangential component of H is continuous:

$$H_2 = H_1 = 10 \text{ A/m}.$$

Step 3: Magnetic Flux Density B_2 in Medium 2

$$B_2 = \mu_2 H_2 = (\mu_0 \mu_{r2}) H_2 = (\mu_0 \cdot 2) \cdot 10 = 20\mu_0 \text{ T}.$$

Step 4: Verify Normal Component of B

The normal components of B are continuous:

$$B_{1n} = B_{2n}.$$

From the calculated values:

- $B_1 = 3\mu_0$
- $B_2 = 20\mu_0$

For B to satisfy continuity, the **angle of the magnetic field vectors** at the boundary will adjust such that the normal components match.

1. Magnetic flux density in each medium:

$$B = \mu H$$

where $\mu = \mu_0\mu_r$ and $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.

2. Relative permeabilities:

- $\mu_{r1} = 0.3$ for medium 1.
- $\mu_{r2} = 2$ for medium 2.

3. Magnetic field intensity:

- $H_1 = 10 \text{ A/m}$ in medium 1.
- Tangential continuity implies $H_2 = H_1 = 10 \text{ A/m}$ in medium 2.

$$B_1 = \mu_1 H_1 = (\mu_0 \mu_{r1}) H_1$$

Substitute $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\mu_{r1} = 0.3$, and $H_1 = 10$:

$$B_1 = (4\pi \times 10^{-7}) \cdot 0.3 \cdot 10$$

$$B_1 = 12\pi \times 10^{-7} \text{ T}$$

$$B_1 = 3.77 \times 10^{-6} \text{ T}.$$

$$B_2 = \mu_2 H_2 = (\mu_0 \mu_{r2}) H_2$$

Substitute $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\mu_{r2} = 2$, and $H_2 = 10$:

$$B_2 = (4\pi \times 10^{-7}) \cdot 2 \cdot 10$$

$$B_2 = 80\pi \times 10^{-7} \text{ T}$$

$$B_2 = 2.51 \times 10^{-5} \text{ T}.$$

The screenshot shows the Scilab 2026.1.0 interface. The console window on the left displays the following text:

```

"Magentic Field Intensity in Medium 1: 10 A/m"
"Magentic Field Intensity in Medium 2: 1.5 A/m"
"Magentic Flux Density in Medium 1: 0.0000038 T"
"Magentic Flux Density in Medium 2: 0.0000038 T"
exec: Wrong number of output argument(s): 0 expected.

Enter magnetic field intensity H1 (A/m) =

```

The editor window on the right shows the following script:

```

1 clear;
2 clear;
3
4 // Relative permeability
5 mu_r1 = 0.9;
6 mu_r2 = 2;
7
8 // Permeability of free space
9 mu_0 = 4 * %pi * 1e-7;
10
11 // Magnetic field intensity in Medium 1
12 H1 = 10;
13
14 // Calculate magnetic field intensity in Medium 2
15 H2 = (mu_r1 / mu_r2) * H1;
16
17 // Calculate magnetic flux density
18 B1 = mu_0 * mu_r1 * H1;
19 B2 = mu_0 * mu_r2 * H2;
20
21 // Display results
22 disp("Magnetic Field Intensity in Medium 1: " + string(H1) + " A/m");
23 disp("Magnetic Field Intensity in Medium 2: " + string(H2) + " A/m");
24
25 disp("Magnetic Flux Density in Medium 1: " + string(B1) + " T");
26 disp("Magnetic Flux Density in Medium 2: " + string(B2) + " T");
27

```

Result:

Thus, the effect of two different media on the magnetic field and flux density across the boundary is calculated.

Parameters	Simulated Result	Theoretical result
1. Magnetic Field Intensity in Medium 1: 10 A/m	Magnetic Flux Density in Medium 1: : 0.0000038 T	Magnetic Flux Density in Medium 1: : 0.0000038 T
2. Magnetic Field Intensity in Medium 2: 1.5 A/m	Magnetic Flux Density in Medium 2: 0.000025 T	Magnetic Flux Density in Medium 2: 0.000025 T